

Foreword

The telecommunications industry is facing major changes with the convergence of telecoms, computing and visual services. Currently we see a number of technologies being applied to meet customer needs with a relatively low level of integration. For example, most corporate users access a range of services, such as the PSTN/ISDN, via their PBXs, the Internet across the corporate LAN, and point-to-point solutions for data transfer. Already we can see a convergence between these technologies with data-based telephony becoming a possibility. In contrast, most residential customers are dependent on the PSTN/ISDN for both telephony and Internet access. For the customers of the cable television vendors there is the possibility of high-speed access, offering the potential of multimedia services operating in real time. However, these services are all supported on different networks that are optimised to the specific services' requirements. None of these networks allows comprehensive interworking between telephony, the Internet and multimedia services, limiting the range and complexity of services that can be offered.

As a result of the convergence in services and the increasing demands they place on bandwidth, the industry has been developing asynchronous transfer mode (ATM) as a single technology flexible enough to satisfy their demands. ATM should support the majority of different user data with a level of control that will allow sophisticated services to be provided. To date ATM switches have been developed and launched and are operational both in the private and public sectors. However, there is still an essential part of the picture missing if ATM is to deliver its full potential. In this issue you will find a series of papers that deal with the network control layer for future ATM-based networks and services. These papers represent an essential part of the broadband picture as they describe the technology required to enable moving from permanent circuits, provided on a point-to-point basis, to switched circuits, allowing a sophisticated service offering. The range of topics covered indicates the importance of broadband to a wide variety of services including future interactive multimedia services, Universal Mobile Telecommunications System and broadband virtual private networks. This special edition covers the topic from a number of perspectives including the collaborative research of Digital Audio-Visual Council and Telecommunications Information Network Architecture Consortium and the standards work of the International Telecommunications Union. Furthermore, there is an in-depth treatment of the technical issues that are currently being studied to allow the timely development of industrial strength products for full service launch. The depth of the technical work presented here positions BT's people among the best in the world and as such is a clear indication of BT's potential to compete successfully in this market. It is clear from the current industry trends that ATM will play an increasingly important role as a high availability, highly reliable and manageable switching technology. It is essential that the industry completes the development of suitable signalling and call control systems, and the associated intelligence, to allow world-class services to be deployed. Once completed, broadband signalling and call control will allow the ATM-based network to carry speech, video and text in a way not possible with current networks. This will allow a new set of multimedia services to be developed and deployed that will truly integrate voice, text and visual services.

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